

# Executive Decision Memo

## Falak Extreme Basmati Rice 5 kg

To: Head of Supply Chain, Sales, Operations, Procurement and Finance

From: Mohammad Omer Farooq, academic demand-planning analysis

Planning snapshot: origin 27 Sep 2026; 12 weeks ending 20 Dec 2026.

**Recommendation.** Plan for 63,478 packs (317.39 tonnes). Make 10,503 packs firm for the first two weeks and keep the remaining 52,980 packs of rounded baseline packing volume adjustable. Rounded weekly quantities total 63,483 packs. Review releases weekly rather than pack the entire horizon immediately.

**Evidence.** Seven models were evaluated on the same unseen 12 weeks. Recursive SMA4 has WAPE 10.26%, MAD 554 packs/week, RMSE 753 packs, average bias -125 packs/week and Tracking Signal -2.72. It improves WAPE by 25.9% relative to naive. Promotion weeks are associated with about 1,530 additional packs in the training regression (HAC  $p < 0.001$ ). Price evidence is inconclusive. These are synthetic associations, not demonstrated causal effects.

**Operating assumptions.** Establish or confirm a 2,500-pack buffer, 6,000-pack weekly capacity and one-week replenishment lead time. Reserve a further 1,000 packs/week as an option. Buffer cost is an illustrative PKR 6 million at assumed cost PKR 2,400/pack. These are planning inputs, not Matco disclosures.

**Risk.** At regular capacity, +20% demand leaves about 3,171 packs unfilled; the reserved capacity option removes this modeled shortage. A one-week delivery interruption still leaves about 2,824 packs unfilled, so secure a backup delivery route or accessible stock. Promotions and availability constraints explain major holdout misses. The holdout contains no Ramadan/Eid weeks and cannot confirm future calendar-event accuracy.

**Action and trigger.** Demand Planning reviews weekly; Sales confirms campaigns; Operations and Procurement confirm stock, capacity and receipts. Reforecast when two weeks' combined actuals differ from their combined forecast by more than 10%, rolling eight-week Tracking Signal exceeds the assumed  $\pm 4$  band, a campaign requires over 6,000 packs/week, or a receipt delay/stockout is confirmed. Activate reserved capacity before known uplift; if unavailable, revise campaign volume or allocation.

This is an academic forward-planning snapshot based on frozen synthetic history. It is not a current operational instruction to Matco.